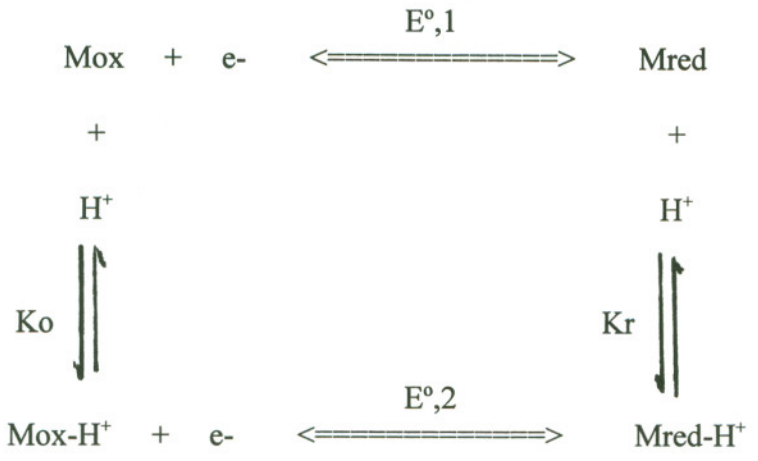
Biochemistry Cume

11/17/00

An electron transfer protein M functions according to the following scheme where the redox potential is dependent on the protonation state of an ionizable group in the protein:



E°,1and E°,2 are the indicated redox potentials for the unrpotonated and protonated forms of M, respectively. Ko and Kr are the <u>association constants</u> for the binding of H^+ to the oxidized and reduced forms of M, respectively.

a. Using this scheme, show that

$$Eh = E^{o},1 + \frac{0.059}{n} \log \frac{(1 + Kr[H+])}{(1 + Ko[H+])}$$

b. How would the expression differ if one wished to express Eh in terms of E°2 rather than E°,1?

c. For the above scheme, sketch a curve showing the dependence of Eh on pH under conditions where pKa,o < pKa,r (the pKa of the ionizable group on the oxidized protein is less than that of the reduced protein). Indicate on your curve, the positions of the relevant pKas and the slope.

d. Show how the pH dependence in part c would differ if pKa,o > pKa,r (the pKa of the ionizable group on the oxidized protein is greater than that of the reduced protein).